AUDITING GENERALIZATION CLAIMS FOR LLM AGENT HARNESSES: SEMANTIC BINDING AND MEASUREMENT VALIDITY

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ABSTRACT

We introduce a *claim-scope framework* for evaluating harness interventions in LLM agents, separating four progressively stronger generalization claims—local effect, within-family held-out confirmation, cross-model transfer, and cross-family transfer—and showing that evidence at one level does not imply the next. On a controlled tool-grounded *semantic-handoff* task, a non-answer-bearing "clarity bundle" (BundleS) suppresses a specific tool-green semantic-binding failure for Qwen3.7-Max (Level 0) and survives a pre-frozen held-out instance (Level 1). Under exact counterbalancing the effect is not established for DeepSeek-V4-Pro (Level 2); cross-family transfer is likewise not established (Level 3): a prospective twelve-instance expansion reversed the descriptive direction relative to a three-instance pilot without establishing a treatment effect, while a second family sits at a non-discriminative ceiling. We introduce a second, orthogonal axis—a four-layer *Harness Effect Audit Protocol* (capability, sampling, execution, artifact integrity)—and show that each layer caught a concrete threat that would have changed the scientific conclusion. An external taxonomy scope probe (Terminal-Bench 2.0→2.1, 26 repairs) shows the taxonomy has external relevance but incomplete coverage (1 direct, 21 partial, 4 outside; sampling layer 0). Our controlled task instances test existence, recurrence, and transfer direction rather than estimate population-level pass rates.

1 INTRODUCTION

When a harness intervention changes agent behavior, what can we claim? We introduce a **claim-scope ladder** with four levels and show that surviving one level does not imply surviving the next:

- **Level 0 (local effect):** a harness intervention changes behavior for a model within a controlled task family.
- **Level 1 (within-family confirmation):** the effect survives a pre-frozen held-out instance from the same semantic/generator family.
- **Level 2 (cross-model transfer):** the effect survives under a different model backend with matched task semantics.
- **Level 3 (cross-family transfer):** the effect survives on independently constructed semantic task families.

Evidence at level k does not imply evidence at level $k+1$. Failure at a higher level does *not* refute the lower-level effect; it bounds the claim's scope.

Our frozen evidence maps to the ladder as follows: Level 0 (Qwen3.7-Max workflow Base 1/3 → BundleS 3/3), Level 1 (pre-frozen held-out: 1/3 → 3/3), Level 2 (DeepSeek-V4-Pro: 3/4 = 3/4, not established), Level 3 (prospective STA n=12: not established; SPICE: ceiling/non-discriminative).

A harness-effect claim has two independent questions: (A) *what scope of generalization does the evidence support?* and (B) *was the measurement itself valid?* The claim-scope ladder addresses (A); a four-layer audit protocol (capability, sampling, execution, artifact integrity) addresses (B). Figure 1 places our concrete evidence and incidents into this 2D framework.

Contributions. (1) A claim-scope framework separating local effects, within-family confirmation, cross-model transfer, and cross-family transfer, with the principle that each level requires independent evidence. (2) A controlled semantic-binding study showing a real local/within-family effect but no established higher-level transfer, including a prospective 12-instance expansion whose descriptive direction reversed relative to a 3-instance pilot. (3) A measurement-validity framework showing how capability, sampling, execution, and artifact failures materially changed scientific interpretation. (4) Supporting external taxonomy evidence from Terminal-Bench repairs.

2 Related Work and Problem Formulation

Harnesses and harness effects. A *harness* is the task-level information structure visible to the agent (prompt, visible files, disclosure bundle, public tool feedback, action surface). A *harness effect* is a change in semantic-binding correctness attributable to harness information content, holding the hidden truth, grader, and tool fixed.

Problem formulation. Each task is a *semantic handoff*: bind a tuple to canonical typed roles from role-misleading evidence. The workflow family binds PVT descriptors to typed axes; Family A (STA/PrimeTime) binds (intent_class, target_partition, check_mode) from an authority provenance DAG; Family B (SPICE/HSPICE) binds (corner, load_condition, metric) from a request–authority join.

TAB vs. semantic binding. Task alignment (Mavali et al., 2026) asks *which* environmental evidence an agent should follow; semantic binding asks *which semantic role* selected evidence is permitted to fill. Our failures can occur after relevant evidence has been retrieved and can remain tool-green—the agent finds the right artifact but binds its value to the wrong typed axis.

Positioning. Yao et al. (2026) broadly characterize model$\times$harness interactions; Wang et al. (2026a) critique harness-leaderboard conflation; Ursekar et al. (2026) optimize harnesses; Zhu et al. (2025) and Shao et al. (2026) address benchmark-construction and protocol-validity standards. We differ by introducing a claim-scope ladder, isolating a controlled semantic-binding mechanism, and auditing which measurement failures changed the scientific conclusion (Table 1). Ma et al. (2026) report cross-benchmark harness gains, which we use as evidence that some interventions do transfer—so transfer must be demonstrated rather than assumed. Concurrently, Liu et al. (2026) audits trajectory-level safety properties of harness execution (boundary compliance, execution fidelity, system stability; 210 tasks across 8 domains); our object differs—we audit the evidentiary scope and measurement validity of semantic-binding performance claims, not execution-trajectory safety.

Table 1: Positioning relative to related work.

Prior work	What it establishes	What this paper additionally does
Harness-Bench (Yao et al., 2026)	Broad model$\times$harness characterization	Claim-scope ladder + controlled semantic mechanism
Rethinking (Wang et al., 2026a)	Leaderboard conflation; search/overfitting	Level k need not imply level $k{+}1$
HarnessOpt-Bench (Ursekar et al., 2026)	Optimizer capability	Studies validity of harness-effect claims
ABC (Zhu et al., 2025) / Protocol validity (Shao et al., 2026)	Setup; shortcut/reward validity	Per-control threat log

3 Methodology: The 2D Framework

The claim-scope ladder (x-axis). Any harness-effect claim lives at one of four levels (Level 0–3 above). Moving right on the ladder requires evidence at the new level; passing a lower level is necessary but not sufficient.

The measurement-validity axis (y-axis). Any measurement has four independent validity layers: (1) capability validity (does the task measure the intended capability, including tool-green semantic alternatives?); (2) sampling validity (repeated trajectories, counterbalancing, membership arbitration); (3) execution validity (transport, terminal-vs-recovered, tool health, protocol completion); (4) artifact integrity (action-surface isolation, canonical-source integrity, custody).

Figure 1: The 2D framework: claim scope (x) $\times$ measurement validity (y). Rows are the four measurement-validity layers (capability, sampling, execution, artifact integrity). Each cell lists the study's concrete evidence or incident.

	Local	Within-family	Cross-model	Cross-family
Cap.	tool-green wrong binding; typed oracle	held-out: BundleS 3/3	DeepSeek 3/4=3/4	STA n=12 not established; SPICE ceiling
Samp.	position-balanced blocking; non-det.	pre-frozen held-out inst.	counterbalanced block	prospective 12-inst. schedule
Exec.	SSE streaming (censoring caught)	recovered degradation kept	—	SPICE action-surface repair
Art.	canonical-tree guard	custody hashes	—	—

Construct: what semantic correctness means. Semantic correctness is *not* inferred from tool success, a hidden numeric answer, or the artifact score. It is decided solely by whether the submitted tuple matches the binding attested by the independent provenance/authority oracle. *Worked example (Family A):* visible evidence attests intent=*functional_close*, partition=*core*, check_mode=*setup*; the agent submits (functional_close, core, *both*)—PrimeTime returns green, the value is type-valid, but the coverage authority attests *setup*, not *both*; the oracle rejects it as a role-conditioned value-selection failure. This construct is instantiated independently in the workflow family (PVT axes), Family A (intent/partition/check_mode), and Family B (corner/load/metric), each with its own grader. We do *not* claim human validation (Study B was preregistered but unexecuted).

A worked instance (workflow family, ambiguous variant). The agent receives an out-of-date timing sign-off handoff and must repair `flow_config.json` to the unique package (netlist, clock, scenario, corner), then regenerate a two-stage evidence chain. The difficulty is deliberately *not* retrieval: the shipped reports label their two axis fields `op_point` and `mode`, and in this variant no glossary, summary or axis declaration is shipped, so which logical axis each label denotes must itself be recovered from the intersection of the sources. Five constraints are recoverable from the specification—C1 netlist family, C2 clock identity (the only clock with non-zero intended-clock coverage), C3 scenario-typed, C4 corner-typed, C5 the sign-off pair. Exhaustive enumeration over the declared domains yields 294 candidate assignments, of which **exactly one** satisfies C1–C5: (`netlist_v2.v`, `clk_main`, `slow`, `func`).

Four evidence artifacts are shipped, each *pairwise plausible* and each violating a different constraint (Table 2). Source A is the critical one: it carries the right netlist *and* the right clock, and **PrimeTime signs it off green**—but its two role fields are swapped, so a corner value occupies the scenario role and a scenario value the corner role. Every shipped source signs off green. No tool signal separates A from the truth; only the typed oracle does. This is what *tool-green* means throughout the paper, and it is why a benchmark that scores this task by tool exit status would score a wrong binding as a pass.

How the oracle adjudicates. The grader never consults tool exit status when deciding semantic correctness. It applies typed-membership predicates to the submitted package: the scenario field must belong to the scenario axis, the corner field to the corner axis, neither may be a PVT descriptor (a descriptor denotes a scenario–corner *pair* and is never itself a value of either axis), and the clock must match by exact identity, so a generic alias is rejected. These predicates are folded into the master evidence gate, so a sign-off-green but mis-typed package cannot pass. A failure is then classified into exactly one of two subtypes, which we never collapse: an **axis-binding failure**, where a value occupies the wrong typed axis (e.g. `func`/`slow`), and a **role-conditioned value-selection failure**, where the value is axis- and type-valid but the wrong member is selected for the role (e.g. `typ`/`func`). Uniqueness of the intended assignment is verified by exhaustive enumeration before an instance is admitted, so "correct" is fixed by the instance's constraint system rather than by grader judgment.

Table 2: The four shipped evidence sources of the worked instance. Each is pairwise plausible, each signs off green under PrimeTime, and each violates a different typed constraint: A and D swap the two role fields, B is out of the netlist family, C puts a PVT descriptor in the corner role and a generic alias in the clock role. The hidden role mapping is `op_point`→scenario, `mode`→corner; it is not disclosed in this variant.

source	netlist	clock	op_point	mode	violates	tool
A role swap	v2 ✓	clk_main ✓	func	typ	C3, C4	green
B stale	v1 ✗	clk_main ✓	slow	func	C1	green
C PVT label	v2 ✓	clk ✗	slow	slow_1.0V_125C	C2, C4	green
D JSON chain	v2 ✓	clk_main ✓	func	typ	C3, C4	green
unique truth	v2	clk_main	slow	func	—	green

Conditions. **Base** (ambiguous handoff; no answer-bearing disclosure); **BundleS** (canonical labels, value-domain definitions, glossary, procedural contract; no answer-bearing C6; no golden values); **TypedContract** (same information as BundleS as a JSON Schema). Hidden truth and grader are identical across conditions.

Design and models. Seeded exact-counterbalanced blocked randomization, frozen before any paid call. Instance is the primary unit; repetitions nested. We evaluate Qwen3.7-Max and DeepSeek-V4-Pro; these are not representative of the frontier-model population. The design is diagnostic, not population-estimating: it tests existence, recurrence, ceiling/null behavior, and transfer direction; it cannot resolve small effect sizes.

4 Study I — Local and Within-Family Effect (RQ1; Levels 0–1)

On the workflow controlled pair (identical truth/grader; ambiguous vs. clear handoff), BundleS suppresses the axis-binding failure for Qwen3.7-Max: Base 1/3 (2 axis-binding) → BundleS 3/3 (0 axis-binding). The schema/contract subset (excluding answer-bearing C6) suffices on the development pair, and generalizes to a *pre-frozen held-out* instance: BundleS 3/3 vs Base 1/3. The failure taxonomy (Figure 2) partitions the workflow ledger (70 episodes) into 41 correct, 24 *axis-binding*, and 5 *role-conditioned value-selection*—all tool-green, rejected only by the typed oracle.

A sequential decomposition tested whether a stable minimal component underlies BundleS. None was isolated: C1 alone, C2-only, C4-only, and a C24 bridge all failed their thresholds. The bundle-level effect is real within-family; the operative sub-component is not identified. Table 3 gives the full per-condition ledger behind both statements.

Three things in Table 3 are worth reading directly. First, the *full* bundle—which does include the answer-bearing component C6—moves both models from failure to 3/3; the scope limitation we report at Level 2 is specifically about the non-answer-bearing subset, not about the phenomenon. Second, the axis column is where the mechanism lives, but it does not localize to one ingredient: every condition that suppresses axis errors carries value-domain information, yet C2-only and C4-only carry it too and still retain three axis failures each, so neither is sufficient in isolation—and the C1-alone row refutes the pre-declared hypothesis that canonical labels alone eliminate axis errors. Third, three condition–model pairs were measured in more than one run window and every one of them disagrees with itself: C24 gives 3/4 with no axis error and then 2/4 with one, Base(0009)/DeepSeek gives 0/3, 2/3 and 3/4 across its three windows, and BundleS/DeepSeek moves from 1/3 to 3/4. That within-condition spread *across* windows, visible only because the ledger does not deduplicate repeats, is the direct evidence for our claim that a component result at k=4 is not stable enough to name a mechanism.

Level 0 and Level 1 are established for Qwen3.7-Max. We make no universal improvement claim. Failure at Level 2 or Level 3 does not refute this local/within-family effect.

Table 3: **Study I episode ledger**, one row per (stage, condition, model): typed-binding correct$/k$, then the two failure subtypes. *Inclusion rule:* all paid, gradeable workflow episodes used in this descriptive ledger, namely the 58 program-primary episodes declared by the frozen manifests of the ablation phases—whose only omissions are the episodes those manifests mark excluded, invalid or aborted (3, 0 and 1)—plus the 12 earlier controlled-pair episodes, which predate that accounting. The 70-episode descriptive ledger is therefore deliberately broader than the 58-episode program-primary accounting reported in the reproducibility statement. A condition measured in two run windows occupies two rows and is never deduplicated—which is why Base(0009)/DeepSeek appears three times and C24 twice. The generating script re-derives every cell from the preserved per-episode submissions and *asserts* each stage's count against the frozen manifest, so a silently dropped stage fails the build instead of shrinking the denominator. Transport is SSE streaming throughout except the DeepSeek controlled-pair rows, which are frozen non-streaming and reported as such.

stage	condition (task)	model	correct$/k$	axis	value
controlled pair	Base(0009)	DeepSeek	0/3	3	0
	Full bundle(0010)	DeepSeek	3/3	0	0
	Base(0009)	Qwen	1/3	2	0
	Full bundle(0010)	Qwen	3/3	0	0
C6 ablation	V1 (0013)	Qwen	3/3	0	0
	V9 (0014)	Qwen	3/3	0	0
screening	BundleD (0016)	Qwen	1/3	2	0
	BundleS (0015)	Qwen	3/3	0	0
held-out	Base(0011)	Qwen	1/3	1	1
	Held(0017)	Qwen	3/3	0	0
cross-model 1	Base(0009)	DeepSeek	2/3	1	0
	BundleS (0015)	DeepSeek	1/3	2	0
cross-model 1C	Base(0009)	DeepSeek	3/4	1	0
	BundleS (0015)	DeepSeek	3/4	1	0
decomp. 1	Contract (0019)	Qwen	1/3	2	0
	Schema (0018)	Qwen	2/3	0	1
decomp. 2	C1 (0020)	Qwen	2/4	2	0
	C24 (0021)	Qwen	3/4	0	1
decomp. 3	C2 (0022)	Qwen	1/4	3	0
	C4 (0023)	Qwen	0/4	3	1
C24 bridge	C24 (0021)	Qwen	2/4	1	1
total			**41/70**	**24**	**5**

Base = ambiguous handoff; Full bundle = C1–C7 incl. answer-bearing C6; V1/V9 = C6 removed from the clear instance / added to the ambiguous one; BundleS = C1+C2+C4+C7 (no C6); BundleD = C3+C5 (decoy disambiguation); Held = held-out instance + BundleS; Schema = C1+C2+C4; Contract = C7 alone; C1 = axis schema; C2 = PVT definitions; C4 = glossary; C24 = C2+C4.

Figure 2: Semantic-binding failure taxonomy.

5 Study II — Cross-Model and Cross-Family Transfer (RQ2; Levels 2–3)

Does the Level 0–1 effect transfer? Table 4 is the main result.

Level 2 (cross-model). Under exact counterbalancing, BundleS is not established for DeepSeek-V4-Pro (3/4 = 3/4). This does not refute the Level 0–1 effect for Qwen3.7-Max; it bounds the claim's scope.

Table 4: Main result matrix (semantic-binding correctness; instance is the unit). Workflow/SPICE entries are correct of k stochastic repetitions; STA entries are the mean correct-rate over 12 instances.

Family / model	Base	BundleS	TypedContract	#inst.	verdict
Workflow / Qwen3.7-Max (Level 0)	1/3	3/3	—	1	local effect
Workflow / Qwen3.7-Max (Level 1, held-out)	1/3	3/3	—	1	within-family
Workflow / DeepSeek-V4-Pro (Level 2)	3/4	3/4	—	1	not established
STA / Qwen3.7-Max (Level 3, prospective)	**0.208**	**0.333**	**0.458**	**12**	**not established**
SPICE / Qwen3.7-Max (Level 3)	1.00	1.00	1.00	3	ceiling

Level 3 (cross-family, STA). Across 12 pre-registered instances, BundleS vs Base: 3 improving, 2 declining, 7 tied; exact sign-test $p = 1.0$; permutation sensitivity $p = 0.31$. *The three-instance pilot and the prospectively frozen twelve-instance batch differed even in their descriptive ordering. We interpret this as evidence that very small task panels are unreliable for directional Harness claims, not as evidence that BundleS transfers; small-panel instability of directional benchmark claims has been independently documented for agent-safety benchmarks (Wang et al., 2026b).* TypedContract is descriptively highest (0.458) but no preregistered TypedContract advantage was established; this is hypothesis-generating only. The pilot (n=3) is not pooled with the prospective n=12.

Table 5: Prospective STA per-instance outcomes (n=12; Y=correct, n=wrong).

	2-rep (rate)			diff		
inst.	Base	BundleS	TC	S−B	TC−B	TC−S
0004	nY(.5)	nn(0)	nn(0)	−.5	−.5	.0
0005	nn(0)	nn(0)	nn(0)	.0	.0	.0
0006	nn(0)	nn(0)	nn(0)	.0	.0	.0
0007	Yn(.5)	nn(0)	nn(0)	−.5	−.5	.0
0008	nn(0)	nn(0)	nn(0)	.0	.0	.0
0009	nn(0)	nn(0)	nn(0)	.0	.0	.0
0010	nY(.5)	YY(1)	YY(1)	.5	.5	.0
0011	nn(0)	YY(1)	YY(1)	1.0	1.0	.0
0012	nn(0)	YY(1)	YY(1)	1.0	1.0	.0
0013	YY(1)	YY(1)	YY(1)	.0	.0	.0
0014	nn(0)	nn(0)	Yn(.5)	.0	.5	.5
0015	nn(0)	nn(0)	YY(1)	.0	1.0	1.0

Level 3 (cross-family, SPICE). Base = BundleS = TypedContract = 1.00—a non-discriminative ceiling. This is a measurement limitation, not evidence against transfer.

Three outcomes, kept distinct. The transfer evidence yields: (i) *no established effect on a discriminating family* (STA, n=12): informative but cannot rule out a small effect; (ii) *ceiling* (SPICE): Base already solves the binding; no headroom; (iii) *direction reversal under prospective expansion* (STA pilot vs. prospective): evidence that very small panels are unreliable for directional claims. We report each separately and do not pool.

6 Study III — Measurement Validity and External Scope Probe (RQ3)

Each headline conclusion could have been manufactured or masked by a measurement artifact. The four-layer protocol is instantiated and stress-tested; Table 6 lists incidents where a control *materially changed the interpretation*.

Table 6: Audit-protocol incidents: each control prevented or corrected an invalid scientific interpretation.

Threat	Control	Conclusion that would have been wrong
Non-streaming censoring (Exec)	SSE + terminal-transport arbiter	Qwen3.7-Max falsely recorded as failing
Position confounding (Samp)	Exact position-balanced blocking	Base/BundleS order leaked into contrast
Recovered degradation (Exec)	Terminal-valid vs recovered split	Hard-but-recovered solve discarded
PT corrupted measurement (Exec)	Tool-health sentinel + bookends	Tool hiccup read as model failure
SPICE action-surface false positive (Art)	Immutable core + forensic audit	12 SPICE episodes spuriously zeroed; semantic-binding ceiling obscured
Canonical-source mutation (Art)	Exact-commit worktree + hash guard	Golden rewrite read as tool outage

Why the controls materially affected interpretation. Without SSE streaming, Qwen3.7-Max would have been recorded as failing the controlled pair—inverting RQ1. Without the SPICE action-surface repair, all twelve SPICE episodes would have been spuriously zeroed, creating an apparent score floor that obscured the true semantic-binding ceiling. Without the canonical-hash guard, a silent golden rewrite would have masqueraded as a day-long tool outage. Each would have altered the headline interpretation.

A monitor is only as trustworthy as the custody of its reference standard. The canonical-source incident deserves a second reading, because its most useful lesson is not that a file was mutated. Our tool-health monitor compares a reference artifact against a frozen fingerprint and reports the tool environment unhealthy on mismatch. During development that monitor fired for roughly a day and the outage was attributed to the remote tool server. The monitor's logic was correct and its mismatch was real; the attribution was wrong, because the artifact it measured against had itself been rewritten—not by the remote server, but by a component of our own test harness that wrote into the canonical tree. A monitor that validates a system against a reference standard therefore inherits the custody of that standard: if the standard can be mutated by the test harness, the monitor can faithfully report a genuine mismatch while misattributing the fault. We state this as a general condition on measurement-validity apparatus rather than as an incident report, and we treat it as a fourth requirement on Layer 4 alongside action-surface isolation, canonical-source integrity and custody: *the reference standards of health checks must be isolated from the harness whose health they certify.* The affected artifact was in the development workspace, which the study already treats as non-authoritative; no reported measurement derives from it. The generalizable point is that this class of error is silent by construction—it presents as a confident, correct-looking alarm about the wrong component.

External taxonomy scope probe (Terminal-Bench 2.0→2.1). As an external scope probe (not validation), we coded the 26 officially documented repairs under the four-layer-plus-Other rubric, *frozen before coding*: **1 direct, 21 partial, 4 outside** (of 26); layers: capability 17, sampling 0, execution 10, artifact 1; 6 multi-layer. We interpret this as evidence that the taxonomy has *external relevance but incomplete coverage*. This is static retrospective taxonomy application; it does *not* validate runtime portability and does *not* validate every audit layer (the sampling layer received no external support). The full 26-task coding is in the Appendix.

7 Limitations and Discussion

Controlled executable tasks, not sampled production incidents. EDA-heavy environments—transfer results are scoped to EDA. *Small instance counts*—Levels 0–2 each rest on single-instance cells at their respective evaluation stage, with Level 1 using a distinct pre-frozen held-out instance; the design is diagnostic, not population-estimating. *Construct validity*—rests on executable provenance/authority oracles; the preregistered human construct-validity study (Study B) is preregistered but unexecuted; we do not claim human validation. *Model scope*—two models, not representative of the frontier population. *Audit protocol*—instantiated and stress-tested here, not broadly validated; the external probe shows relevance but incomplete coverage. *No mechanism isolated*—the bundle-level effect is real within-family but no stable minimal component was identified.

What would change these conclusions. Because the framework's whole purpose is to make a claim's scope contestable, we state in advance what evidence would move each verdict; none of it requires trusting our interpretation.

- *Level 0–1 (established here) would be weakened or revised* by a preregistered counterbalanced replication on the same frozen instances that fails to reproduce the Base–BundleS separation for Qwen3.7-Max, or by a demonstration that the BundleS text leaks the answer—i.e. that a solver could recover the unique assignment from the bundle without consulting the evidence. The bundle was constructed to exclude the answer-bearing component precisely so this is checkable: the frozen condition files are released, and C6 is separable.
- *Level 2 (not established) would be resolved either way* by raising k on DeepSeek-V4-Pro under the same exact counterbalancing. Our 3/4 = 3/4 cannot distinguish a genuinely absent effect from an effect too small for k=4; we claim only the former is unestablished, not that it is absent.
- *Level 3 (not established on STA) could be resolved toward transfer* by a prospectively frozen panel with enough power to distinguish a preregistered, practically meaningful effect from zero. We do not treat the local Level 0–1 result as implying a cross-family effect size; a within-family effect entails nothing about its magnitude elsewhere, which is why the panel's target must be preregistered on its own grounds. Our n=12 with 7 ties has little power; the honest reading is a wide interval around zero, not a demonstrated null.
- *The ceiling result (SPICE) would become informative* under a harder instance generator for the same family—one where Base does not already solve the binding. A ceiling is a property of our instances, not of the family.
- *The mechanism claim (none isolated) would be revised* by any single component reproducing the axis-suppression pattern across two independently collected run windows. The two C24 rows of Table 3 are exactly this test, and they disagree.
- *The audit protocol's coverage claim would be strengthened or refuted* by applying the frozen rubric to a second external repair corpus. Our single probe found the sampling layer entirely unsupported, which is evidence of incomplete coverage that a second corpus could confirm or overturn.

Two results are, by contrast, not contestable by re-analysis alone, because they are properties of the measurement rather than of the models: the transport-censoring incident and the action-surface false positive would have produced confident wrong numbers in any analysis of the uncorrected logs. That asymmetry is the argument for reporting the audit layer alongside the result.

Discussion. The claim-scope ladder makes generalization claims auditable: a reader can locate any harness-effect claim at a specific level and ask whether the evidence supports it. The measurement-validity axis makes the measurement independently checkable. Together they answer: *what scope does the evidence support, and was the measurement itself valid?* Some harness interventions do transfer (Ma et al., 2026); transfer is therefore an empirical property that must be demonstrated, not assumed.

8 Conclusion

We introduced a claim-scope framework for harness-effect generalization claims and showed that a local/within-family effect (Levels 0–1) did not establish cross-model (Level 2) or cross-family (Level 3) transfer. A prospective 12-instance expansion reversed the descriptive direction while still not establishing a treatment effect. The measurement-validity framework showed that capability, sampling, execution, and artifact controls each materially changed the scientific conclusion. The contribution is a framework for auditing what evidence supports a generalization claim—not a proposal of an improved harness.

REFERENCES

Chengzhi Liu, Yichen Guo, Yepeng Liu, Yuzhe Yang, Qianqi Yan, Xuandong Zhao, Wenyue Hua, Sheng Liu, Sharon Li, Yuheng Bu, and Xin Eric Wang. Auditing agent harness safety. In *arXiv preprint*, 2026. arXiv:2605.14271.

Ziyu Ma, Hailang Huang, Shun Zou, Yong Wang, Shidong Yang, Yiming Hu, Fei Wei, and XiangXiang Chu. LongHorizon-Harness: Advancing long-horizon agents for real-world tasks. In *arXiv preprint*, 2026. arXiv:2608.01964.

Sina Mavali, David Pape, Jonathan Evertz, Samira Abedini, Devansh Srivastav, Thorsten Eisenhofer, Sahar Abdelnabi, and Lea Schönherr. No more, no less: Task alignment in terminal agents. In *arXiv preprint*, 2026. arXiv:2605.12233.

Jiaqi Shao, Hanck Chen, Wei Zhang, Maxm Pan, and Bing Luo. Do agent benchmarks measure capability? protocol validity in the age of agentic AI. In *arXiv preprint*, 2026. arXiv:2607.22368.

Varun Ursekar, Apaar Shanker, Yash Maurya, Shehab Yasser, Vijay S. Kalmath, Veronica Chatrath, and Yuan Xue. HarnessOpt-Bench: Evaluating LLMs at harness optimization. In *arXiv preprint*, 2026. arXiv:2608.06301.

Yike Wang, Huaisheng Zhu, Zhengyu Hu, Yige Yuan, Zhengyu Chen, Shakti Senthil, Hannaneh Hajishirzi, Yulia Tsvetkov, Pradeep Dasigi, and Teng Xiao. Rethinking the evaluation of harness evolution for agents. In *arXiv preprint*, 2026a. arXiv:2607.12227.

Youting Wang, Xiao Han, Dingyan Shang, Yuan Tang, and Bowen Liu. Safety, or just capability? a validity audit of agent-safety benchmarks. In *arXiv preprint*, 2026b. arXiv:2607.28685.

Yilun Yao, Xinyu Tan, Chao-Hsuan Liu, Yaoming Li, Zhengyang Wang, Wenhan Yu, Zhewen Tan, Yuxuan Tian, Guangxiang Zhao, Lin Sun, Xiangzheng Zhang, and Tong Yang. Harness-Bench: Measuring harness effects across models in realistic agent workflows. In *arXiv preprint*, 2026. arXiv:2605.27922.

Yuxuan Zhu, Tengjun Jin, Yada Pruksachatkun, Andy Zhang, Shu Liu, Sasha Cui, Sayash Kapoor, Shayne Longpre, Kevin Meng, Rebecca Weiss, Fazl Barez, Rahul Gupta, Jwala Dhamala, Jacob Merizian, Mario Giulianelli, Harry Coppock, Cozmin Ududec, Jasjeet Sekhon, Jacob Steinhardt, Antony Kellermann, Sarah Schwettmann, Matei Zaharia, Ion Stoica, Percy Liang, and Daniel Kang. Establishing best practices for building rigorous agentic benchmarks. In *arXiv preprint*, 2025. arXiv:2507.02825.

A STUDY I — PER-INSTANCE LEDGER AND MINIMAL-COMPONENT ABLATION

The full per-condition ledger (Table 3) now appears in the main text. Every cell is re-derived from the preserved per-episode records (`flow_config.submitted.json`, `result.json`, sanitized agent log) by a committed script and is typeset directly from that script's output, so the table cannot drift from the records by transcription. Program-wide: 70 episodes (the 58 program-primary episodes of the ablation phases plus the 12 earlier controlled-pair episodes), 41 correct, 24 axis-binding, 5 role-conditioned value-selection. The script also enforces the inclusion rule mechanically—it asserts each stage's episode count against the frozen program manifest and aborts on mismatch—because an earlier aggregation of these same records keyed cells by (condition, model) in a dictionary and so silently dropped repeat measurements of a condition; that is a failure mode a stated rule alone does not prevent. Minimal-component ablation (exploratory): no stable sub-component isolated—C1 alone, C2-only, C4-only and the C24 bridge all failed their pre-declared thresholds.

B STUDY II — PROSPECTIVE STA DETAIL

The per-instance table (Table 5) appears in the main text. Historical pilot (n=3) not pooled.

Statistical procedure (frozen; descriptive sensitivity only). The task instance (n=12) is the independent unit; two repetitions are nested observations (72 trajectories are *not* treated as n=72). Pilot instances 0001–0003 are excluded from the primary contrast. Each per-instance, per-condition rate is the mean over its two repetitions. The primary contrast is the signed per-instance difference $d_i = \mathrm{BundleS}_i - \mathrm{Base}_i$. *Exact paired sign test.* Of the 12 instance differences, 5 are non-zero (3 positive, 2 negative; 7 ties), so the test is conditioned on n=5 non-zero differences with k=3 positive; the two-sided exact p-value is $2\sum_{i=0}^{\min(k,n-k)}\binom{n}{i}/2^n = 1.0$, excluding ties (standard convention). This is a non-establishment result, not evidence of a zero effect. *Per-instance permutation sensitivity.* Under the sharp null of no condition effect, we reshuffle the six within-instance condition labels (2 Base, 2 BundleS, 2 TypedContract) independently per instance, recompute $\sum_i(\mathrm{BundleS}_i - \mathrm{Base}_i)$, and report the two-sided proportion of 10^4 Monte-Carlo replicates (fixed seed 20260811) whose absolute re-randomized sum meets or exceeds the observed $\sum_i d_i = 1.5$, giving $p = 0.31$. This is a descriptive sensitivity check over the fixed instance set, *not* a population-rate p-value (the instances are not sampled from a defined population). No adaptive k was used; no trajectory-pooled p-value is reported; results are reported at the preregistered analysis.

C STUDY III — FULL 26-TASK TERMINAL-BENCH REPAIR CODING

Rubric frozen before coding; source: PR #53 "Terminal-Bench 2.1" (body sha256 `c7172fc6`; files sha256 `fccf2d61`); 2.0/2.1 snapshots frozen (89 tasks each; sha256 `5c7ed05c`).

Task	threat	mechanism	coverage
adaptive-rejection-sampler	Cap	instruction clarification	partial
build-pmars	Cap	instruction clarification; verifier/test tightening	partial
caffe-cifar-10	Exec; Cap	timeout; verifier/test broadening; instruction; env/dep; resource	partial
compile-compcert	Exec	environment/dependency repair	partial
configure-git-webserver	Cap	oracle repair; verifier/test tightening	partial
extract-moves-from-video	Exec	external-dependency removal	partial
filter-js-from-html	Cap; Exec	instruction clarification; resource increase	partial
fix-git	Art	hidden-artifact removal; verifier/test tightening	direct
hf-model-inference	Cap	verifier/test broadening	partial
install-windows-3.11	Cap	instruction clarification; verifier/test tightening	partial
make-doom-for-mips	Exec	oracle repair; env/dep repair	partial
mteb-leaderboard	Exec; Cap	env/dep; instruction; resource	partial
mteb-retrieve	Exec; Cap	env/dep; instruction clarification	partial
polyglot-c-py	Cap	verifier/test broadening	partial
polyglot-rust-c	Cap	verifier/test broadening	partial
protein-assembly	Cap	oracle repair	partial
rstan-to-pystan	Cap	oracle repair	partial
sam-cell-seg	Cap	instruction clarification	partial
torch-tensor-parallelism	Cap; Exec	instruction clarification; resource increase	partial
torch-pipeline-parallelism	Exec	resource increase	partial
query-optimize	Cap	instruction clarification	partial
train-fasttext	Exec; Cap	verifier/test tightening; env/dep repair	partial
build-pov-ray	Other	env/dep repair	not-covered
financial-document-processor	Other	env/dep repair	not-covered

mcmc-sampling-stan	Other	env/dep repair	not-covered
overfull-hbox	Other	env/dep repair	not-covered

Aggregate: direct 1 / partial 21 / not-covered 4; Cap 17, Exec 10, Art 1, Other 4; Samp 0; 6 multi-layer.

D OPERATIONAL INDEPENDENCE AND PHASE CHRONOLOGY

Five pre-registered structural criteria (independent templates, vocabularies, truths, graders, decoys). Historical chronology (provenance; does not structure the main paper): discovery → transport repair → controlled pair → held-out → cross-model → cross-family → TypedContract → prospective STA. Episodes 58/24/36/72; costs ¥682.25 / ¥7.72 / ¥11.75 / ¥43.56. The 58 is the workflow program-primary accounting declared by the frozen ablation-phase manifests; the 70-episode descriptive Study I ledger of Table 3 is broader because it additionally includes the 12 earlier controlled-pair episodes, which predate that accounting and its cost ledger.

E INFRASTRUCTURE + CUSTODY

Exact-commit worktree; canonical-hash guard (`FAILED_INTEGRITY` stop); episode arbiter; SSE streaming; tool-health sentinel; immutable-core/derived-deck repair; per-episode custody byte-matching. Pre-flight caught a grading-path bug (malformed signoff JSON) before the prospective run; historical pilot unaffected (0/30 triggers).

Ethics statement. The preregistered blinded human construct-validity study (Study B) is preregistered but unexecuted because qualified independent human annotators were unavailable; no LLM annotator was substituted.

AI-use statement (outside the page limit). Generative-AI assistants were used to support experimental design and critique, orchestration and infrastructure code, task generators, analysis and report scripts, literature search, result-interpretation checks, and manuscript drafting and editing. All experimental decisions, task definitions, acceptance criteria, statistical analyses, and scientific claims were reviewed and approved by the authors. Experimental results were produced by frozen executable pipelines and independently checkable graders reproduced from frozen manifests; generative-AI systems were not used as ground-truth judges or scientific oracles. The authors take responsibility for all content and conclusions.

Reproducibility statement. Deterministic generators (fixed seeds); independent per-family graders; exact-commit isolated-worktree execution with pre/post-episode canonical-hash verification; committed sample-membership arbitration; validity-only replacement; per-episode custody byte-matching. Program totals: 58+24+36+72 paid primary episodes; committed-ledger cost ¥745.29. (The descriptive Study I ledger of Table 3 reports 70 workflow episodes: these 58 plus the 12 earlier controlled-pair episodes, which sit outside this program accounting and its cost ledger.) Frozen manifests, randomization schedules, custody hashes, per-episode sanitized evidence, and the scripts that re-derive every table from the preserved records are in the anonymized supplement. ICLR 2027 deadlines: abstract September 18, 2026 AOE; full paper September 25, 2026 AOE.